MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, LOGIC POWER SUPPLY

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Logic Power Supply Circuit Card Assembly: 9355846, used in the Mortar Ballistics Computer (MBC): M23.Througout this specification this assembly is referenced as "supply".

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MILI"STD-109	Quality Assurance Terms and Definitions

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-454	-	Electronic Equipment, General Requirements for
MIL-STD-1521	-	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STD-45662	-	Calibration System Requirements

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Goverment documents, drawings, and publications. The following other Government documents, drawings, and publications-form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal.

DRAWINGS

9355846	Circuit Card Assembly, Logic Power Supply
11785850	Computer Set, Ballistics, Mortar: M23

PACKAGING DATA SHEETS

9355846	Packaging of Circuit Card Assembly, . Power Supply Logic: 9355846

(Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

Item definition. The logic power supply circuit card assembly shall provide the Mortar Ballistic Computer (MBC) with logic level supply voltages from various battery voltages or from unregulated external prime power. It also provides necessary power up/down signals, low battery indicator, and automatic switchover from prime power to battery sources. The connector contacts are listed in Table I.

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TABLE I. Logic power supply circuit card assembly connector contacts.

Connector	Pin	Type	Name
J1	Al		+28VR
J1	A2	I	+28V
J1	B1		+28VR
J1	B2	I	+78V
P1	1		BAT+5V
P1	2	0	-5V
P1	3		GND
P1	4		GND
P1	5	I	BATPOS
P1	6	I	BATPOS
P1	7	0	+5V
P1	8	0	+5V
P1	9		GND
P1	10		GND
P1	11		GND
P1	12	0	RESET*
P1	13	0	PSFAULT*
P1	14	0	BATLOW
P1	15		OFF
P1	16		ELHI
P1	17		ON*
P1	18		ELON
P1	19		ELLO
P1	20		DSPON*
P1	21		+35V
P1	22		MC1200
XA2P1	Al		+35V
XA2P1	A2		ELHI
XA2P1	A3		ELLO
XA2P1	A4		MC1200
XA2P1	Bl		ELON
XA2Pl	B2		DSPON*
XA2P1	B3		ON*
XA2P1	B4		OFF
XA2P1	Al	0	PS+5V
XA2P1	A2	0	+5V
XA2P2	A3	0	SPOS
XA2P2	A4	I	PSRUN*
XA2P2	B1		BAT+5V
XA2P2	B2		GND
XA2P2	B3		GND
XA2P2	B4	0	PSFAULT*

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3.2 Performance.

3.2.1 General. The performance of the supply shall be in accordance with the following requirements. Logic levels shall be as specified in Table II.

TABLE II. Logic levels.

Type of IC Logic	Logic '0'	Logic '1'
CMOS "and HCMOS (Vcc = 5.3V)	0.5 +/- 0.5 volts DC	4.6 +/- 0.7 volts DC

3.2.1.1 Power supply run. The supply shall ture on when XA2P2-A4 is connected to P1-9 (GND) and-shall turn off when disconnected. It shall be considered on when requirement 3.2.1.3 is met.

3.2.1.2 Negative volts. P1-2 shall be minus 5.3 plus or minus 0.2 volts DC when referenced to GND.

3.2.1.3 Positive volts. P1-7 shall be plus 5.3 plus or minus 0.2 volts DC when referenced to GND.

3.2.1.4 Output ripple. Both the +5V output (P1-7) and the -5V output (P1-2) shall have an AC component with a valtage level less than 0.2 volts peak to peak that is at a frequency less than 1 megahertz.

3.2.1.5 Overload/shortcircuit protection. The +5V output (Pi-7) shall begin to decrease in voltage when the output current is at least 560 milliamperes and at most 900 milliamperes. When the +5V output is shortcircuited to GND, the output current shall be less than 1.7 amperes.

3.2.1.6 Reset. During turn on conditions, PI-12 shall stay at logic O for at least 150 milliseconds and at most 500 milliseconds after P1-13 (PSFAULT*) switches from at logic 0 to a logic1. Then it will switch to a logic 1. During turn off conditions, P1-12 shall stay a logic 1 for at least 100 microseconds and at most 500 microseconds after PSFAULT* switches from a logic 1 to a logic O. Then it will switch to a logic O.

3.2.1.7 Power supply fault. P1-13 shall be at logic O if P1-7 (+5V) is less than 4.9 volts DC and shall be a logic 1 if it is greater than 5.1 volts DC.

3.2.1.8 Power supply volts. XA2P2-A1 shall be 5.6 plus or minus 0.3 volts DC when referenced to GND.

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3.2.1.9 Supply positive. XA2P2-A3 shall be 9 volts DC minimum when referenced to GND with BATPOS at 10 volts DC. XA2P2-A3 shall be 30 volts DC minimum when referenced to GND with +28V at 32 volts DC.

3.2.1.10 Battery low. PI-14 shall be at logic 1 if P1-5 (BATPOS) is less than 10.7 volts DC and shall be at logic O if it is greater than 11.3 volts DC.

3.2.1.11 Common connector contact continuity. The resistance between the following respective connector contacts shall be less than 7.5 ohms.

J1-A1	to	J1-B1	P1-16	to	XA2P1-A2
J1-A2	to	J1-B2	P1-17	to	XA2P1-A3
P1-l	to	XA2P2-B1	P1-18	to	xA2P1-Bl
P1-3	to	P1-4	P1-19	to	XA2P1-A3
P1-4	to	P1-9	P1-20	to	XA2P1-B2
P1-5	to	P1-6	P1-21	to	xA2P1-Al
P1-7	to	P1-8	P1-22	to	XA2P2-A4
P1-8	to	XA2P2-A2	XA2P2-B2	to	XA2P2-B3
P1-9	to	P1-lo			
P1-lo	to	P1-11			
P1-11	to	XA2P2-B2			
P1-13	to	XA2P2-B4			
P1-15	to	XA2P2-B4			

3.2.2 Physical characteristics. The physical characteristics shall be as prescribed by the drawings, specifications, and QAP's-

3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques employed in the fabrication of the supply shall not degrade the inherent reliability. The inherent reliability shall not be less than 285,243 hours Mean Time Between Failures (MTBF). (See 4 and 6)

3.3 Environmental. The supply shall meet the requirements specified herein unless otherwise specified.

3.3.1 Room ambient temperature. The requirements of this specification shall be met with the supply thermally stabilized at room ambient temperature of +16°C to +32°C (+60°F to +90°F), unless otherwise specified in the requirements ParagraPh.

3.3.2 Environmental stress screen (ESS) temperature. The supply shall meet the requirements of 3.2.1.1 and 3.2.1.11 inclusive, after having been exposed to a minimum of ten temperature cycles at temperatures of -57°C and +71°C (-71°F and +160°F).

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3.3.3 Environmental stress screen (ESS) vibration. The supply shall meet the requirements of 3.2.1.1 and 3.2.1.11 inclusive, after having been subjected to a random or pseudorandom vibration which includes frequency content from 100 Hz to 1000 Hz and up to a 3g RMS in two axis for a period of 10 minutes plus or minus 15 seconds.

3.4 Fabrication. The supply shall be manufactured in accordance with drawing 9355846.

3.5 First article. When specified in the contract, a sample shall be subjected to first article inspection (See 4 and 6).

3.6 workmanship. The workmanship requirements of MIL-F-13926 and MIL_STD_454 shall be met. All components shall be fitted properly and firmly in their respective positions. All electricial connections shall be mechanically secure and electrically sound. There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely affect its use or suitability for the purpose intended.

3.7 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as . specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) supplies plus three (3) each of all items covered by QAP. The three supplies shall be tested as specified herein for all the requirements of this specification.

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The items covered by QAP shall be tested, as specified thereon, for all characteristics delineated on the QAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor prior to submitting-the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reseves the right to witness any or all of the contractor's inspections.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provision of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected In accordance with the inspection provisions contained in Quality Assurance Provisions (QAP's) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4 Sampling.

4.4.1 Environmental/performance. The supply shall be inspected to the requirements in Table III.

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TABLE III. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
100% Inspection			
1.	Power supply run	3.2.1.1	4.6.3.1
2.	Negative volts	3.2.1.2	4.6.3.2
3.	Positive volts	3.2.1.3	4.6.3.3
4.	Output ripple	3.2.1.4	4.6.3.4
5.	Overload/shortcircuit protection	3.2.1.5	4.6.3.5
6.	Reset	3.2.1.6	4.6.3.6
7.	Power supply fault	3.2.1.7	4.6.3.7
8.	Power supply volts	3.2.1.8	4.6.3.8
9.	Supply positive	3.2.1.9	4.6.3.9
10.	Battery low	3.2.1.10	4.6.3.10
11.	Common connector contact continuity	3.2.1.11	4.6.4
12.	ESS temperature	3.3.2	4.6.1
13.	ESS vibration	3.3.3	4.6.2
MAJOR: AQL 1.0% Defective			
101.	Workmanship	3.6	4.6.5

4.4.2 Special sampling. One supply shall be selected at random as a special sample from each lot or each 100 produced which ever occurs first. The sample shall meet the requirements and tests in Table IV.

TABLE IV. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
201.	Fabrication	3.4	Applicable Drawing\ Visual
202.	Materials	3.7	Applicable Drawing/ Visual

NOTE: The tests in Table IV shall be conducted at room ambient temperature (as specified in 3.3.1).

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

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4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-4507.

4.5.1 Government furnished inspection equirement. Where the contract provides for Government furnished test equirement, supply and maintenance of test equirpment shall be in accordance with the applicable requirements specified in MIL-1-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number in specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List (TDPL).

4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply inspection equipment compatible with the “Test Methods and Prodecures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream,whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against ratilation effects.

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4.5.2.3.2 Fabrication, setup and calibration. fabrication of test equipment shall be such that it incorporates the features of rigidity, stability? and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The supply shall be placed in the test chamber and the temperature of the chamber reduced 10°C (18°F) per minute plus or minus 10 (1.8°F) from room ambient temperature (as specified in 3.3.1) to -57°C (-71°F). The supply shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the chamber shall be raised 10°C (18°F) per minute plus or minus 10 (1.8°F) to +71°F (+160°C). The supply shall remain at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (18°F) per minute plus or minus l°C (1.8°F) to room ambient temperature (as specified in 3.3.1). The aabove procedure shall be conducted a minute of ten times. Upon completion of the cycles the supply shall then be removed from the chamber and subjected to the tests in 4.6.3.1 and 4.6.4 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.11 inclusive.

4.6.2 ESS vibration. Conduct the vibration test with the supply positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. Unless otherwise specified, the test equipment shall be in accordance with the “Test Facilities” requirements of MIL-F-13926. The applied motion shall be in two separate planes for the time period specified. Upon completion of the vibration sweep the supply shall be removed from the vibration adapter and shall be subjected to the tests in 4.6.3.1 and 4.6.4 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.11 inclusive.

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4.6.3 Performance. The performance tests 4.6.3.1 and 4.6.3.10 inclusive, shall be conducted with a 21 plus or minus 11 volts DC power supply connected to J1-A2 with respect to J1-Al, a 1 kilohm plus or minus 5 percent 1/4 watt resistor connected between Pi-2 and GND, a 91 ohm plus or minus 5 percent 1/2 watt resistor connected between P1-7 and GND, and a 100 kilohm plus or minus 5 percent 1/4 watt resistor between P1-12 and P1-7, a second one between P1-13 and P1-7, and a third between P1-14 and P1-7.

4.6.3.1 Power supply run. Connect XA2P2-A4 to GND. The supply shall turn on as specified in 3.2.1.1.

4.6.3.2 Negative volts. Turn on the supply while monitoring P1-2. The output response shall be as specified in 3.2.1.2.

4.6.3.3 Positive volts. Turn on the supply while monitoring P1-7. The output response shall be as specified in 3.2.1.3.

4.6.3.4 Output ripple. Turn on the supply. With an oscilloscope, look at the output ripple at P1-7 and P1-2, both with respect to GND. The output response shall be as specified in 3.2.1.4.

4.6.3.5 Overload/shortcircuit protection. Disconnect the 91 ohm resistor from P1-7 and GND. Connect a variable O to 10 ohm 10 watt resistor in series with a 5 amp dc current meter. Connect the variable resistor current meter combination between P1-7 and GND. Set the resistor to its maximum resistance and turn on the supply. Slowly decrease the resistance while at the same time noting the current meter reading and the DC voltage level at P1-7 with respect to GND. The output response shall be as specified in 3.2.1.5. ReconneCt the 91 ohm resistor between P1-7 and GND.

4.6.3.6 Reset. Turn on the supply while monitoring P1-12 and P1-13. The output response shall be as specified in 3.2.1.6.

4.6.3.7 Power supply fault. Turn on the supply. Note the voltage at P1-7 during the time P1-13 switches from a logic O to a logic 1. The output response shall be as specified in 3.2.1.7.

4.6.3.8 Power supply Volts. Turn on the supply. The output response shall be as specified in 3.2.1.8.

4.6.3.9 Supply positive. Turn on the supply.The output response shall be as specified in 3.2.1.9.

4.6.3.10 Battery low. Disconnect the DC power supply from JI-A1, connect a 15 volt DC variable power supply to PI-5 with respect to GND, and turn on the supply. Slowly decrease the 15 volt power supply until P1-14 switches from a logic O to a logic 1. Note the voltage level of the applied voltage at the time P1-14 swiches. The output response shall be as specified in 3.2.1.10.

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4.6.4 common connector contact continuity. This test shall be conducted with no external power connected to the supply. Measure with an ohm meter the resistance between the connector contacts specified in 3.2.1.11. The resistances shall be as specified in 3.2.1.11.

4.6.5 Workmanship. The supply shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The supply shall conform to the requirements of 3.6.

4.7 Reliability testing. The requirement for reliability test approval and the responsibility (Government or contractor) for reliability testing shall be as specified in the contract. The sample for reliability test shall consist of 3 supplies. The samples shall be manufactured in the same manner, using the same materials, equirpment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts. and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.7.1 Government testing. When the Government is responsible for conducting reliability tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample~ including certificates of conformance for all materials.

4.7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspected and testted by the contractor for all the requirements of the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractors reliability tests and inspections.

5. PREPARATION FOR DELIVERY

5.1 Delivery requirements. Handling, packing packing shipping, and marking of the Power Supply Logc Circuit Card Assembly, 9355846 shall be in accordance with Packaging Data Sheet 9355846. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The power supply logic circuit card assembly described herein is intended for use in the MBC. See drawing 11785850.

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6.2 Ordering data. procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d, Applicable Packaging Data Sheet.

e. Applicable levels of preservation, packing and marking (see 5.1).

f. Applicable stock numbers.

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